

Ann Mary Thomas

Data Analyst | Energy & Sustainability Analyst

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PROFILE

Data, energy, and sustainability analyst with a background spanning mechanical engineering, data analytics, and AI/ML modelling. Currently at WMG, University of Warwick, embedded within a manufacturing SME (3T Additive Manufacturing) as part of the government-funded STRATA project - delivering carbon footprint reduction strategies, building Power BI dashboards, and developing data-driven models to drive operational efficiency. Research experience in deep learning-based energy forecasting at London Metropolitan University's Genesis Lab, with two journal papers on AI modelling and energy field impacts currently under review. Proven ability to turn complex datasets into clear, actionable insights for technical and non-technical stakeholders across industry and academia.

CORE SKILLS & TECHNOLOGIES

Data Analysis	Python (Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow), SQL, Excel (Advanced), Statistical Analysis
AI & Machine Learning	LSTM, RNN, Attention Mechanisms, Hybrid Deep Learning, Predictive Modelling, Time-Series Forecasting
Visualisation & BI	Power BI, Dashboard Design, Data Storytelling, KPI Reporting, HTML UI Development
Energy & Sustainability	Carbon Footprinting, Scope 1/2/3 Emissions, CO2 Forecasting, Decarbonisation Roadmaps, Energy Consumption Analysis, CII/IMO Standards
Systems & Tools	SAP ERP, ERP/MRP/MES awareness, Git, Jupyter Notebook, Data Pipeline Automation, Data Cleansing & Validation
Professional Skills	Stakeholder Engagement, Technical Report Writing, Cross-team Collaboration, Grant Application Support, Research & Publication

PROFESSIONAL EXPERIENCE

Energy Consumption & Sustainability Analyst

Dec 2025 – Present

WMG, University of Warwick - 3T Additive Manufacturing (STRATA Project)

Government-funded project embedding a specialist analyst within SME manufacturer 3T Additive Manufacturing to drive energy optimisation and decarbonisation through data.

- Design and maintain Power BI dashboards integrating manufacturing process data to enable real-time monitoring of energy KPIs, CO2 emissions, and operational efficiency metrics.
- Build and automate data pipelines for energy and carbon data collection, cleansing, and validation - ensuring consistent, high-quality inputs for reporting and decision-making.
- Develop carbon footprint reduction roadmaps, conducting scenario analysis and cost-benefit assessments to prioritise and evidence decarbonisation interventions.
- Work with SAP ERP systems to extract and structure manufacturing data for sustainability analysis and reporting.
- Produce technical reports and presentations for stakeholders, project partners, and government funding bodies - translating complex data into clear, actionable narratives.
- Collaborate with engineering and operations teams to gather data requirements and implement evidence-based sustainability improvements.

Energy Consumption & Sustainability Analyst (Intern)

Sep 2025 – Dec 2025

3T Additive Manufacturing

- through systematic energy data analysis and targeted operational interventions across manufacturing processes. Achieved a 39.3% carbon footprint reduction
- Collected and analysed manufacturing energy consumption data to identify patterns, inefficiencies, and optimisation opportunities.
- Defined and tracked Scope 1 and 2 emissions KPIs; built interactive Power BI dashboards for real-time sustainability monitoring and management reporting.
- Authored technical reports translating analytical findings into management recommendations, directly informing strategic decisions.

Research Assistant (Part-time)

Feb 2025 – Present

London Metropolitan University - Genesis Lab

- Independently designed and implemented a Hybrid AI architecture (LSTM + Attention) to forecast electricity demand and CO2 emissions, in collaboration with Dr. Maitreyee Dey.
- Conducted scenario-based analysis quantifying CO2 reductions achievable through renewable energy integration - findings being prepared for publication.

- Two journal papers on AI modelling and its effects in the energy field currently under review for peer-reviewed academic publication.

Junior Data Analyst

Aug 2022 – Jan 2024

Navalt Solar and Electric Boats

- Analysed operational data for 600+ vessels (100,000+ records), building predictive models to evaluate fuel efficiency and estimate CO2 emissions using engine performance and sea state variables.
- Developed an internal HTML-based UI tool for fuel consumption prediction, enabling non-technical engineering staff to run and interpret model outputs directly.
- Automated data workflows using Python scripts, reducing preparation time by 75% and improving pipeline reliability and reporting accuracy.
- Independently researched IMO Carbon Intensity Indicator (CII) metrics and applied them to fleet emissions analysis, expanding expertise in maritime carbon accounting and environmental compliance.
- Collaborated with engineering and sustainability teams to align analytical insights with decarbonisation goals and client reporting requirements.

RESEARCH & PUBLICATIONS

- **Two journal papers on AI modelling and its effects in the energy field - currently under peer review.**
- Energy demand & CO2 forecasting project: implemented and benchmarked LSTM, RNN, and RBM+NN models on hourly US power sector data (California & Texas). Optimised LSTM achieved ~5% improvement on MAE/RMSE over baseline. Highlights peak-demand forecasting improvements and decarbonisation implications of renewable integration.
- Student volunteer - FICTA'25 (Frontiers of Intelligent Computing: Theory and Applications), international conference.

SELECTED PROJECTS

Vehicle Emissions Power BI Dashboard

- Built interactive Power BI dashboard analysing CO2 emissions across vehicle types (EU environmental dataset). Identified engine size as accounting for ~70% of emission variance; implemented multi-dimensional interactive filters for manufacturer, country, and fuel-type analysis.

Graduate Outcomes Analysis - Machine Learning (Python)

- Applied Random Forest classification and K-means clustering to UK HESA education data; identified four graduate profiles used to recommend targeted university support programmes.

Marketing Campaign Response Model (Python)

- Built logistic regression model on 1,500 customer records to predict campaign response; identified key demographic features that improved targeting accuracy by 20%.

EDUCATION

MSc in Data Analytics

2024 – 2025

London Metropolitan University

B.Tech in Mechanical Automobile Engineering

2017 – 2021

SCT College of Engineering

KEY ACHIEVEMENTS

- 39.3% carbon footprint reduction at 3T Additive Manufacturing through data-driven sustainability analysis and targeted decarbonisation interventions.
- Two AI energy forecasting papers under review - advancing research into CO2 reduction through renewable energy integration.
- Reduced data processing time by 75% via Python automation at Navalt Solar and Electric Boats.
- Independently designed LSTM + Attention hybrid model architecture for electricity demand and CO2 forecasting.
- Invited into a full WMG analyst role following internship performance at 3T Additive Manufacturing.

CERTIFICATIONS

- Excel for Data Analytics - Simplilearn
- Statistics for Data Science - Simplilearn